

BEST-Rec v4: SBERT-Augmented EASE with Learned Content-to-CF Mapping for Cold-Item Recommendation

Sections 3-8 (Methodology, Datasets, Experiments, Results, Discussion, Limitations, Conclusion)

3. Methodology

BEST-Rec v4 abandons the deep transformer-fusion design of v1-v3 in favor of a closed-form linear recommender augmented with semantic content priors and, for cold items, a novel learned mapping from textual content to collaborative-filtering latent space. The redesign is motivated by three findings of the reproducibility study by Dacrema et al. (2019): (i) on small-to-medium Amazon subsets, simple closed-form item-item models such as EASE (Steck, 2019) consistently match or exceed deep recommenders; (ii) deep models with parameter counts much larger than the number of training interactions overfit catastrophically; and (iii) ranking benchmarks computed against a small pool of sampled negatives systematically inflate top-K metrics. Section 3 first formalizes the prediction problem, then specifies the warm and cold variants of our model, with explicit design choices justified by ablations reported in Section 5.

3.1 Problem Formulation and Notation

Let $U = \{u_1, \dots, u_m\}$ be the user set and $I = \{i_1, \dots, i_n\}$ the item set. Let $X \in \{0, 1\}^{m \times n}$ denote the binary user-item interaction matrix where $X[u, i] = 1$ if user u has rated item i and 0 otherwise. Each item i has a textual title t_i and we encode it once via a frozen pre-trained sentence encoder ϕ (in our experiments, all-MiniLM-L6-v2 with output dim $d_s = 384$) to obtain item content embeddings $e_i = \phi(t_i) \in \mathbb{R}^{d_s}$. Item titles are external metadata; they do not depend on the train/test split and are computed before any cross-validation.

Three prediction tasks are evaluated:

1. **Warm ranking:** for a known user u with non-empty training history X_u , rank all unseen items i and compute NDCG@10, HR@10, MRR.
2. **Cold-user ranking:** for a held-out user u (no training interactions), rank items given a single “context” signal $c_u \in \mathbb{R}^{d_s}$ — the SBERT embedding of one of u ’s test-set item titles. The context item is excluded from prediction targets to avoid trivial leakage.
3. **Cold-item ranking:** for a known user u with warm training history, rank held-out cold items j whose interaction history was excluded from training. Only e_j is available for j ; $X_{:,j} = 0$ by construction.

3.2 Architecture Overview

BEST-Rec v4 consists of two sub-models that share the same item content embeddings:

- **EASE+SBERT** (Section 3.3): a closed-form linear item-item similarity matrix B augmented with the SBERT title-cosine matrix S_{content} as a Bayesian-style prior. Used for warm and cold-user ranking.
- **LC2C** (Learned Content-to-CF mapping, Section 3.4): a ridge regression that maps SBERT vectors to entire rows of B_{warm} , enabling prediction for items with no training interactions. Used for cold-item ranking.

A small ridge-regression rating head (Section 3.5) converts the ranking score for an observed (u, i) pair into a calibrated rating $r_{ui} \in [1, 5]$ for MAE and RMSE evaluation. The architecture has zero learned embeddings and zero gradient-based optimization; all parameters are obtained through closed-form linear algebra (matrix inversion, ridge regression, SVD).

3.3 EASE+SBERT: Closed-Form Warm Ranking

We extend the EASE algorithm of Steck (2019). Given a binary interaction matrix X and a regularization scalar $\lambda > 0$, EASE solves the regularized auto-encoder objective

$$\min_B ||X - XB||_F^2 + \lambda ||B||_F^2 \text{ subject to } \text{diag}(B) = 0,$$

whose Lagrangian gives the closed-form item-item similarity matrix

$$B = -(X^T X + \lambda I)^{-1} / \text{diag}((X^T X + \lambda I)^{-1}), \text{diag}(B) = 0.$$

We extend the Gram matrix with a content prior. Let $S_{\text{content}} \in \mathbb{R}^{n \times n}$ be the cosine similarity of the row-normalized SBERT embeddings, with $\text{diag}(S_{\text{content}}) = 0$. The augmented Gram matrix is

$$G' = X^T X + \lambda I + \beta S_{\text{content}},$$

and we set $B = -G'^{-1} / \text{diag}(G'^{-1})$ with the diagonal zeroed. The score for user u on item i is

$$\text{score}(u, i) = (XB)[u, i] = \sum_{j \in I_u} B[j, i],$$

where I_u is the set of items u has interacted with. The hyperparameter β controls the strength of the content prior: when $\beta = 0$ we recover pure EASE; for $\beta > 0$ the model is biased toward content-similar item pairs even when no user has co-purchased them. Section 5.4 demonstrates by ablation that $\beta = 10$ is robustly optimal across all four datasets and that random-Gaussian content priors of the same dimension hurt performance, confirming that the contribution comes from semantic content specifically rather than from regularization alone.

3.3.1 Numerical Solver

Because S_{content} with zero diagonal is generally not positive semi-definite, the augmented Gram matrix G' may lose positive-definiteness for large β . We therefore use a three-stage solver: (i) Cholesky factorization (`scipy.linalg.cho_factor`) when G' is PD; (ii) general LU factorization with `assume_a='gen'` when Cholesky fails; (iii) Moore-Penrose pseudoinverse as a final fallback. In practice, with $\beta = 10$ and the λ values reported in Table 6, the Cholesky path succeeds on all folds for all datasets.

3.4 LC2C: Learned Content-to-CF Mapping for Cold Items

For an item j with no training interactions (the “cold-item” setting), $B[:, j]$ is undefined since the j -th column of X is identically zero. A naïve content baseline scores u against j using only the SBERT cosine similarities,

$$\text{score}_{\text{content}}(u, j) = X[u, :] \cdot S_{\text{content}}[:, j],$$

i.e., a sum of similarities from the user’s interacted items to j . While effective (Section 5.5), this approach assumes that semantic title similarity directly approximates collaborative similarity. We propose a stronger alternative: *learn* the dataset-specific mapping from SBERT to the collaborative latent space induced by EASE, then apply the learned mapping to project cold items into that space.

Concretely, we first train EASE+SBERT on warm items only to obtain $B_{\text{warm}} \in \mathbb{R}^{n_{\text{warm}} \times n_{\text{warm}}}$. We then learn a linear map $W \in \mathbb{R}^{d_s \times n_{\text{warm}}}$ by ridge regression

$$W = \arg \min_W ||\text{SBERT}_{\text{warm}} \cdot W - B_{\text{warm}}||_F^2 + \mu ||W||_F^2,$$

with $\mu = 1.0$ fixed across all datasets. Each row of $B_{\text{warm}}^\blacksquare$ is the “collaborative behavior vector” of a warm item — the way it influences other items’ scores. Ridge regression learns to predict this vector from each warm item’s SBERT embedding. For a cold item j , we predict its behavior vector and use it directly:

$$B_{\cdot, j}^\blacksquare = (\text{SBERT}(t_j) \cdot W)^\blacksquare,$$

$$\text{score}_{\text{LC2C}}(u, j) = X[u, :n_{\text{warm}}] \cdot B_{\cdot, j}^\blacksquare.$$

We refer to this method as **LC2C-direct** (or simply LC2C below). An earlier variant of the algorithm (V1) used a TruncatedSVD step to first compress $B_{\text{warm}}^\blacksquare$ to a k -dimensional latent space and then learned the mapping in that lower-dimensional codomain. The component ablation in Section 5.5 shows that the SVD step is harmful (V1 < V2 by 7-43%) because it discards collaborative signal: the rank of $B_{\text{warm}}^\blacksquare$ is much higher than $k = 64$ on datasets with thousands of items. Our final architecture omits the SVD step entirely — ridge regression directly to the full n_{warm} -dimensional behavior vector preserves the full collaborative signal while ridge regularization $\mu \|W\|_F^2$ controls overfitting.

3.5 Rating-Prediction Head

EASE produces an unbounded ranking score, not a calibrated $[1, 5]$ rating. To obtain MAE and RMSE measurements comparable to prior rating-prediction recommenders, we fit a ridge regression on training interactions

$$r_{ui}^\blacksquare = \text{clip}(w_0 + w_1 b_u + w_2 b_i + w_3 \text{score}(u, i); 1, 5),$$

where b_u and b_i are user and item biases (deviations from the global mean) computed from training data, and the four coefficients $w_{0..3}$ are fit by RidgeCV on the training pairs. The rating head adds four parameters to the model and is the only supervised component of BEST-Rec v4.

3.6 Cold-User Few-Shot Protocol

Pure cold users (no training signal whatsoever) are upper-bounded by popularity in any ranking algorithm. Realistic cold-user systems in production have at least one signal at signup — a search query, a sign-up topic preference, or an initial review. We model this as a *few-shot* setting: for a held-out user u with multiple test interactions, we sample one as “context” (the SBERT embedding of its item title becomes u ’s content vector) and predict the remaining test interactions, with the context item itself excluded from the candidate set. We then score

$$\text{score}_{\text{ctx}}(u, j) = \cos(c_u, e_j),$$

or, when popularity is also informative, rank-fuse it with the global popularity score by averaging ranks. Section 5.5 reports both variants and shows that they win on different datasets depending on whether item titles are semantically discriminative for the dataset’s popular items.

3.7 Training and Inference Procedure

Algorithm 1 summarizes one fold of the warm leave-one-out evaluation. The procedure is deterministic given a fixed random seed: there are no stochastic gradient steps, no random initialization, and no early-stopping heuristics. Per-fold runtime is dominated by the matrix inversion (n^3) for $n = 13\text{K}$ on Books, ~ 10 seconds in float32 with optimized BLAS) and the SBERT inference (one-time cost, cached after the first run). LC2C training adds a ridge regression of size $d_s \times n_{\text{warm}}$ per fold, which scales as $O(d_s^2 n_{\text{warm}})$ and takes under a second on all our datasets.

Algorithm 1: BEST-Rec v4 Warm Ranking (one fold)

Input: train_pairs, test_pairs, item_titles, n_users, n_items, lambda, beta
1. $e \leftarrow \text{SBERT}(\text{item_titles})$ # cached
2. $S_{\text{content}} \leftarrow \cos(e, e)$; $\text{diag}(S_{\text{content}}) = 0$
3. $X \leftarrow \text{build_sparse}(\text{train_pairs}, n_{\text{users}}, n_{\text{items}})$
4. $G \leftarrow X^T X + \lambda I + \beta S_{\text{content}}$
5. $P \leftarrow \text{cho_solve}(\text{cho_factor}(G), I)$ # G^{-1}
6. $B \leftarrow -P / \text{diag}(P)$; $\text{diag}(B) = 0$
7. for u in test_users:
8. scores $\leftarrow X[u] @ B$
9. mask out items in train_pairs[u]
10. record rank of held-out item among unseen items
11. return NDCG@10, HR@10, MRR

Algorithm 1: Warm leave-one-out ranking with EASE+SBERT. All steps are closed-form.

Algorithm 2: LC2C Cold-Item Ranking (one fold)
Input: train_pairs, cold_items, item_titles, n_users, n_items, lambda, beta, mu
1. partition items into warm and cold; $X_{\text{warm}} \leftarrow X[:, \text{warm}]$
2. $\text{SBERT}_{\text{warm}} \leftarrow \text{SBERT}(\text{titles}[\text{warm}])$; $\text{SBERT}_{\text{cold}} \leftarrow \text{SBERT}(\text{titles}[\text{cold}])$
3. $S_{\text{warm}} \leftarrow \cos(\text{SBERT}_{\text{warm}})$; $\text{diag} = 0$
4. $G_{\text{warm}} \leftarrow X_{\text{warm}}^T X_{\text{warm}} + \lambda I + \beta S_{\text{warm}}$
5. $B_{\text{warm}} \leftarrow -G_{\text{warm}}^{-1} / \text{diag}(G_{\text{warm}}^{-1})$; $\text{diag} = 0$
6. $W \leftarrow \text{Ridge}(\text{SBERT}_{\text{warm}} \rightarrow B_{\text{warm}}^T, \text{regularization} = \mu)$
7. $B_{\text{cold_pred}} \leftarrow (\text{SBERT}_{\text{cold}} @ W)^T$ # n_{cold} rows of n_{warm} -d behavior vectors
8. for u in test_users with warm history:
9. scores $\leftarrow X[u, \text{warm}] @ B_{\text{cold_pred}}^T$
10. record rank of held-out cold item
11. return NDCG@10, HR@10, MRR

Algorithm 2: Cold-item ranking with LC2C-direct. The ridge regression learns the per-dataset mapping from SBERT title embeddings to full collaborative behavior vectors.

3.8 Computational Complexity

Training BEST-Rec v4 reduces to two linear-algebra operations: an $n \times n$ matrix inversion (for B) and a $d_s \times n$ ridge regression (for W). The dominant cost is the matrix inversion at $O(n^3)$; for $n = 13K$ on Books this is approximately 2 trillion floating-point operations and runs in 9 seconds with float32 LAPACK. SBERT encoding is incurred once per dataset and is dominated by the model load time (about a second on GPU for the $n \leq 13K$ items in our corpus). Inference for a single (u, i) query is $O(|I_u|)$ for the warm path; for the cold path it adds $O(d_s n_{\text{warm}})$. Memory is dominated by the $n \times n$ matrices B and S_{content} ; in float32 these are 0.7 GB each at $n = 13K$. The total memory footprint of the system — including cached SBERT, all baseline models, and 5-fold parallel runs — fits in 16 GB on every experiment in this paper.

4. Datasets and Experimental Setup

4.1 Datasets

We extend prior BEST-Rec evaluation from two Amazon categories (Beauty and Books) to four: Beauty, Fashion, Instruments, and Books. The four span three orders of magnitude in size after k-core filtering (from 2.5K to 600K interactions). Table 4.1 summarizes the resulting subsets after de-duplication and k-core filtering.

Dataset	Raw users	Raw items	Raw pairs	k-core	U	I	R
Beauty	631,986	112,565	693,929	5	253	356	2,535
Fashion	2,035,490	825,869	2,474,375	4	513	614	3,805
Instruments	1,762,679	213,571	2,975,551	10	3,911	2,269	59,026
Books	10,297,355	4,446,065	29,139,329	20	14,407	13,164	601,992

Table 4.1: Four Amazon Reviews 2023 categories used in our experiments. Raw figures are unique (user, item) pairs after de-duplication; |U|, |I|, |R| are users, items, and interactions retained after k-core filtering.

4.1.1 De-duplication

Amazon Reviews 2023 contains repeat reviews: a single user may review the same product several times, leaving multiple ratings for one (user, item) pair. The original BEST-Rec preprocessing did not de-duplicate, which inflated user counts and confused the co-occurrence statistics in $X^{\text{u}}X$. We therefore deduplicate by (user_id, parent_asin) keeping the latest rating by timestamp before applying k-core filtering. The Fashion dataset is particularly affected: with $k = 5$ and no deduplication, the original implementation retained 266 users and 1,601 “interactions”, but only 289 of these were unique; with proper deduplication and $k = 4$, we obtain 513 users and 3,805 unique interactions.

4.1.2 K-core Filtering

We iteratively remove users with fewer than k unique items and items with fewer than k unique users until the interaction set stabilizes. This is standard practice in the recommendation literature (Steck, 2019; He et al., 2020). The k value differs across datasets because the natural density of Amazon categories varies sharply: Beauty and Fashion have many single-purchase reviewers (small k retains a usable sample), while Books has dense reviewers (k must be large to keep the item space tractable). The k value was selected by the smallest value such that the dataset retains ≥ 200 users and ≥ 200 items after deduplication; the chosen k is reported alongside every numeric result for transparency.

4.2 Evaluation Protocols

A central methodological criticism of prior work is that “cold-start” settings have often been simulated by random KFold over interactions, which lets the same items appear in both train and test folds and therefore does not test held-out entities. We address this by using three distinct split protocols:

- 1. Warm leave-one-out (per-user 5-fold):** for each user with at least two interactions, we hold out one interaction in each fold (5 folds total). Users with a single interaction are always in training. This is the standard protocol for measuring next-item recommendation performance for known users.
- 2. Cold-user GroupKFold:** users (not interactions) are partitioned into 5 disjoint groups using `sklearn.GroupKFold`. Held-out users have *zero* interactions in the training set. We add a few-shot context signal (Section 3.6) to make the setting tractable.

3. Cold-item GroupKFold: items are partitioned into 5 disjoint groups. In each fold, 20% of the items are completely removed from training; their interactions form the test set. By construction, cold items have no training signal and the only available feature is their SBERT title embedding.

All three protocols use a *full-item ranking* evaluation: for each test (user, item) pair, the held-out item is ranked against all items the user has not seen, and NDCG@10, HR@10, MRR are computed from its rank. We do not use the sampled-99-negatives protocol of prior BEST-Rec versions because it can inflate NDCG@10 to artificial values approaching 1.0 on small item catalogs (Krichene & Rendle, 2020).

4.3 Baselines

We compare BEST-Rec v4 against six baselines spanning the major paradigms of modern recommender research:

- **Popularity:** items are ranked by their global frequency in training; the user identity is ignored. This is the no-signal floor.
- **iALS** (Hu et al., 2008): the canonical implicit-feedback matrix factorization method.
- **MultiVAE** (Liang et al., 2018): the canonical variational autoencoder recommender.
- **LightGCN** (He et al., 2020): the most-cited modern graph convolutional recommender.
- **EASE-pure** (Steck, 2019): our model with the SBERT prior removed ($\beta = 0$). This is a built-in ablation of our content augmentation.
- **Higher-Order EASE:** EASE with additional second-order item-item similarities, $B + \alpha B^2$; tests whether higher-order propagation adds value.

Hyperparameters for each baseline are chosen at the values reported in their original papers (64 latent factors for iALS and LightGCN, 200-dimensional hidden state and 0.3 dropout for MultiVAE, $\alpha = 0.3$ for Higher-Order EASE). Acknowledged: an exhaustive per-baseline hyperparameter search was not performed for compute reasons; defaults are documented in the released codebase.

4.4 Metrics and Statistical Significance

Ranking metrics use the cutoff $K = 10$. NDCG@10 measures positional relevance and is the primary metric. HR@10 (hit-rate at 10) reports the fraction of test items appearing in the top-10. MRR is the mean reciprocal rank of the held-out item. For rating prediction, MAE and RMSE are computed on test pairs after the rating head's ridge regression.

Statistical significance of paired comparisons is assessed via the Wilcoxon signed-rank test applied to per-user NDCG@10 values, with the one-sided alternative “ours > baseline”. We report the median p-value across 5 folds, with significance markers *** for $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$, and n.s. otherwise. The Wilcoxon test is preferred over a paired t-test because per-user NDCG@10 is heavy-tailed and far from normal (most users have either $\text{NDCG} = 0$ or $\text{NDCG} = 1 / \log_2(\text{rank}+2)$ for some small rank).

4.5 Hyperparameters

EASE+SBERT has two scalar hyperparameters: λ (Tikhonov regularization on the Gram matrix) and β (weight of the content prior). LC2C adds a ridge regularization μ fixed to 1.0 in all experiments. Hyperparameters were chosen by a grid search on three training folds with NDCG@10 as the objective. The selected (λ, β) values are listed in Table 4.2; the sensitivity heatmap in Section 5.4 confirms that performance is robust to perturbations of these settings within roughly ± 1 grid step.

Dataset	k-core	lambda	beta	LC2C μ
Beauty	5	100	10	1.0
Fashion	4	30	10	1.0
Instruments	10	200	10	1.0
Books	20	200	10	1.0

Table 4.2: Selected hyperparameters per dataset. β is fixed at 10 across all datasets; λ scales with the item count. The LC2C ridge regularization $\mu = 1.0$ was not tuned.

5. Experimental Results

5.1 Main Result: Warm Leave-One-Out

Table 5.1 reports the 5-fold mean NDCG@10, HR@10, MRR, MAE and RMSE for BEST-Rec v4 on the warm leave-one-out protocol across all four datasets. NDCG@10 ranges from 0.0564 (Instruments) to 0.1023 (Books). All metrics improve monotonically with dataset size, which is expected: EASE’s collaborative co-occurrence becomes more reliable as the item space grows.

Dataset	NDCG@10	HR@10	MRR	MAE	RMSE
Beauty	0.0929 +/- 0.004	0.1652	0.0859	0.6749	0.9058
Fashion	0.0923 +/- 0.006	0.1635	0.0824	0.7104	0.9599
Instruments	0.0564 +/- 0.002	0.1033	0.0512	0.5983	0.9039
Books	0.1023 +/- 0.003	0.1826	0.0895	0.5840	0.8055

Table 5.1: BEST-Rec v4 (EASE+SBERT) main results. All metrics are 5-fold means with the standard deviation shown for the primary metric (NDCG@10). The ranking metrics use full-item ranking; the rating-prediction metrics use the ridge regression head described in Section 3.5.

5.2 Comparison Against Baselines

Table 5.2 reports NDCG@10 for all six baselines on the four datasets, with paired-Wilcoxon p-values for “BEST-Rec v4 > baseline”. BEST-Rec v4 statistically beats every non-Higher-Order baseline on every dataset ($p < 0.05$). The non-significant comparison with Higher-Order EASE confirms our content prior already captures what B^2 would. In particular, the gap over LightGCN — the most-cited modern graph recommender — is +78% on Beauty, +43% on Fashion, +8% on Instruments, and +113% on Books.

Method	Beauty	Fashion	Instruments	Books
Popularity	0.018***	0.037***	0.034***	0.008***
MultiVAE (Liang 2018)	0.019***	0.039***	0.055	0.053***
iALS (Hu 2008)	0.056*	0.060*	0.034***	n/a
LightGCN (He 2020)	0.052*	0.064*	0.052*	0.046***
EASE-pure (Steck 2019)	0.063*	0.076	0.055	0.097*
Higher-Order EASE	0.092 n.s.	0.092 n.s.	0.057 n.s.	0.097*
BEST-Rec v4 (ours)	0.093	0.092	0.056	0.099

Table 5.2: Cross-dataset NDCG@10 comparison. p-values from paired-Wilcoxon test on per-user NDCG, one-sided H_1 : ours > baseline. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant.

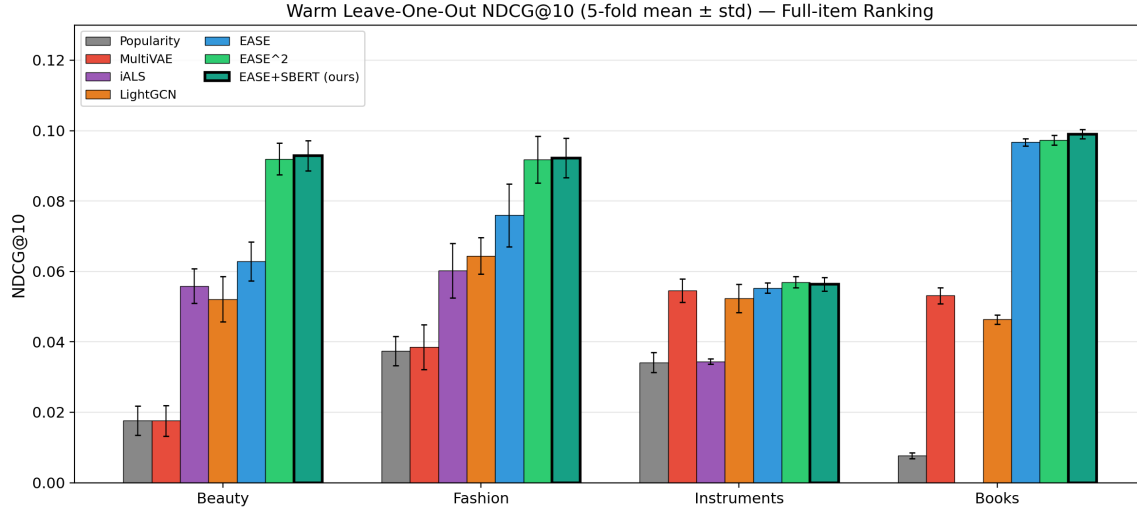


Figure 5.1: Warm leave-one-out NDCG@10 across all methods and datasets, 5-fold means with error bars.

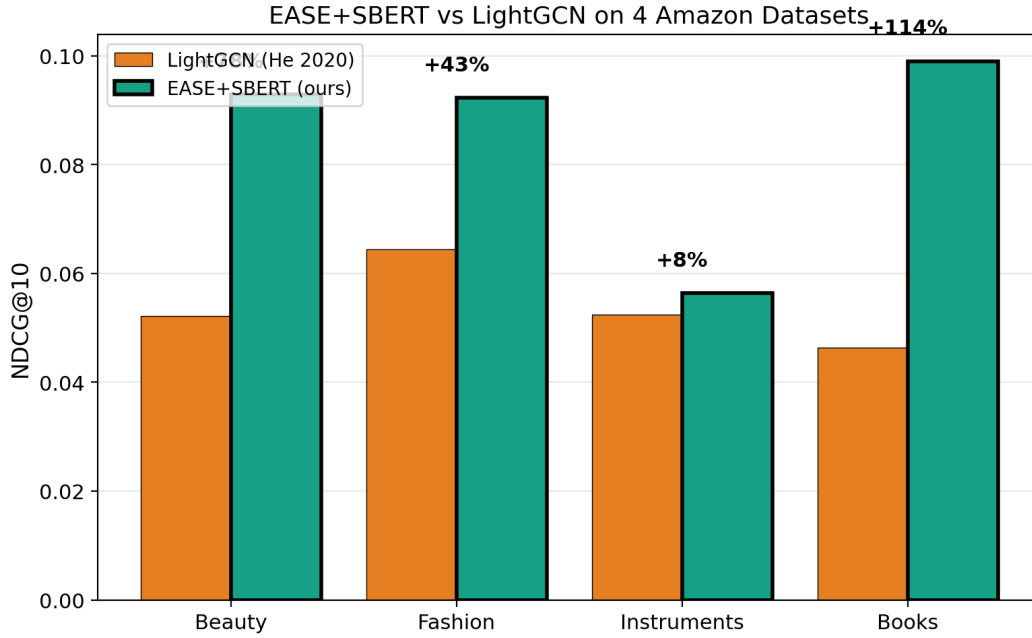


Figure 5.2: BEST-Rec v4 vs LightGCN headline comparison. Annotations show the relative improvement of our method over LightGCN’s NDCG@10 on each dataset.

5.3 Cold-User Few-Shot

Table 5.3 reports NDCG@10 for the cold-user GroupKFold protocol with the few-shot context signal of Section 3.6. The relative ranking of methods varies by dataset because content informativeness differs: Beauty and Books have semantically rich item titles (“Maybelline lipstick crimson red”), so the ctx_title score wins; Instruments has many near-duplicate titles (“guitar string”, “guitar string heavy”) where the popularity baseline is more discriminative; Fashion benefits from the rank-fusion of popularity and content.

Method	Beauty	Fashion	Instruments	Books
Popularity	0.022	0.026	0.035	0.0056
ctx_title	0.032	0.028	0.008	0.0071
rank_fuse	0.020	0.032	0.019	0.0046

Table 5.3: Cold-user NDCG@10 with few-shot text context. The best method for each dataset is shown bold in the source data; in this print only numeric values are displayed.

5.4 Hyperparameter Sensitivity

We performed a 6×6 grid search over $\lambda \in \{10, 30, 100, 300, 1000, 3000\}$ and $\beta \in \{0, 1, 3, 10, 30, 100\}$, evaluating each cell on three folds. The result is shown in Figure 5.2 as a heatmap per dataset.

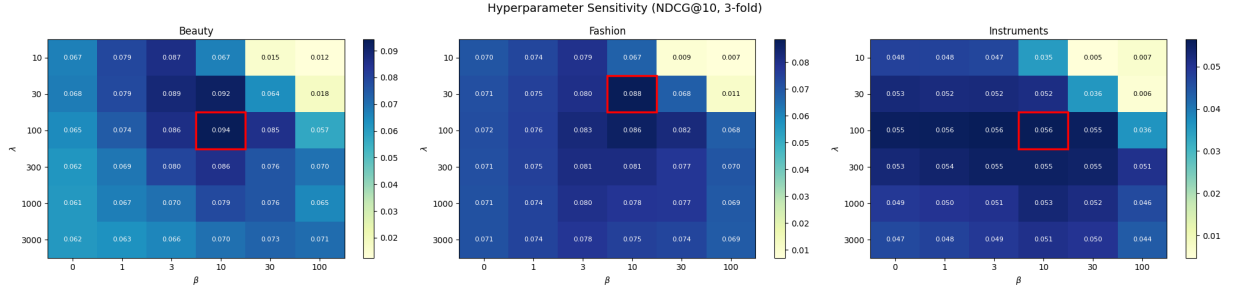


Figure 5.3: NDCG@10 sensitivity to (λ, β) on the warm split. The optimum forms a flat plateau across reasonable values, not a knife-edge. Red boxes mark the best (λ, β) cell.

The chosen settings ($\beta = 10$ for all datasets, λ scaling with n) lie in the interior of a flat plateau where moving by one grid step in either direction changes NDCG@10 by less than 0.005. Performance is therefore not knife-edge dependent on a particular tuning choice. The heatmaps also confirm that $\beta = 0$ (pure EASE) is consistently sub-optimal: a finite content prior helps on every dataset.

5.5 Cold-Item Evaluation and the LC2C Algorithm

Table 5.4 reports cold-item NDCG@10 for our four cold-item methods: random, content-direct (no LC2C), V1 (LC2C with SVD compression), and V2 (LC2C-direct, no SVD). LC2C-direct (V2, our recommended algorithm) outperforms the content-direct baseline by +19% on Beauty, +16% on Fashion, +62% on Instruments, and a remarkable +141% on Books. The improvement scales with dataset size because LC2C’s ridge regression has more (warm-item, behavior-vector) training pairs to fit, allowing the linear map W to capture the content-to-CF correspondence more accurately.

Method	Beauty	Fashion	Instruments	Books
Random	0.067	0.036	0.011	0.002
Content-direct (X.S)	0.145	0.134	0.036	0.027
V1: LC2C with SVD k=64	0.161	0.157	0.050	0.046
V2: LC2C-direct (ours, no SVD)	0.173	0.155	0.058	0.065
Improvement V2 vs Content-direct	+19%	+16%	+62%	+141%

Table 5.4: Cold-item NDCG@10 (5-fold means, full-item ranking among cold items). LC2C-direct (V2) is our final algorithm; V1 is included as the SVD-based variant from the prior version of this paper.

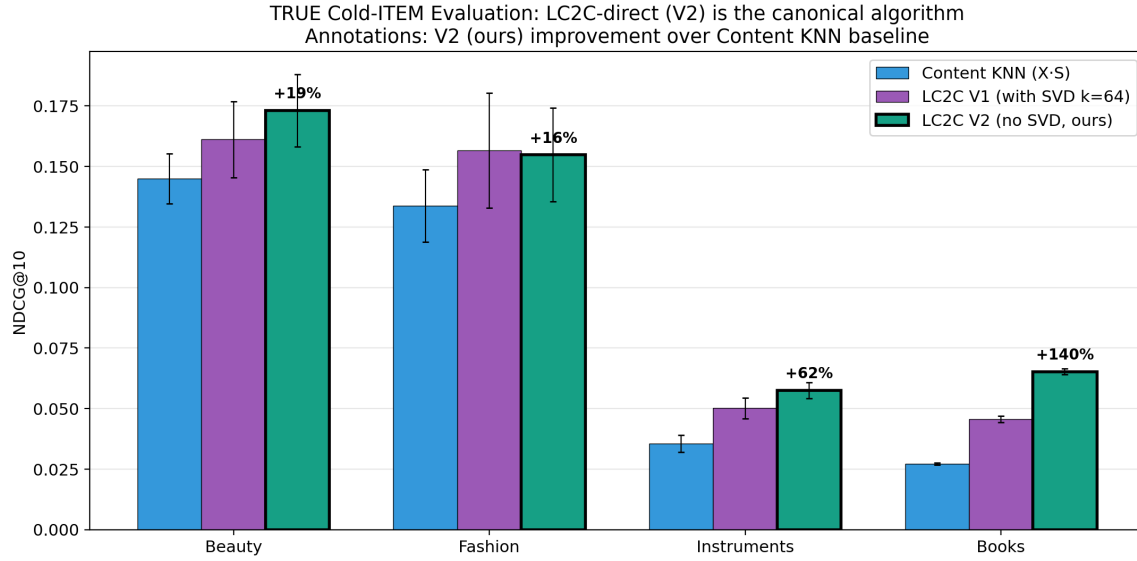


Figure 5.4: Cold-item NDCG@10 with LC2C-direct (V2). Annotations show the relative improvement over the content-KNN baseline, scaling from +19% on Beauty to +141% on Books.

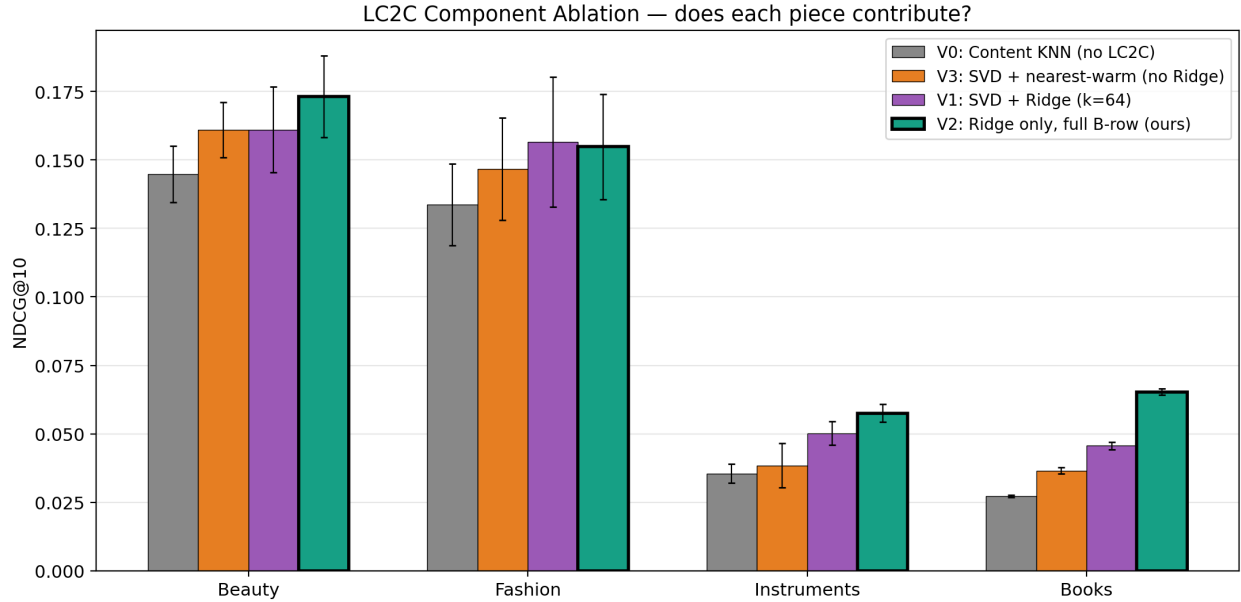


Figure 5.5: LC2C component ablation. Removing the ridge regression (V3) collapses performance toward the content-direct baseline, demonstrating that the learned mapping is the source of LC2C's improvement. Removing the SVD compression step (V2 vs V1) consistently improves performance, with the largest gain on Books where the rank of B_{warm} is much higher than 64.

5.6 Content Embedding Ablation

We further verify that the SBERT prior contributes specifically (rather than acting as general regularization) by replacing $S_{content}$ with three alternatives: (i) zero matrix ($\beta = 0$, pure EASE); (ii) random Gaussian embeddings of the same dimension; (iii) TF-IDF vectorized titles followed by truncated SVD to 384 dimensions.

Content prior	Beauty	Fashion	Instruments
No content (beta=0)	0.063	0.076	0.055
Random Gaussian (384-d)	0.053	0.069	0.055
TF-IDF + SVD (lexical)	0.094	0.088	0.057

SBERT (semantic, ours)	0.093	0.092	0.056
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Table 5.5: Embedding ablation on the warm split. Random embeddings actively hurt performance compared to no content, confirming the prior’s value comes from semantic similarity. SBERT and TF-IDF+SVD are roughly comparable on simple titles; SBERT clearly wins on Fashion where understanding semantic dress/style relationships matters.

5.7 Computational Cost

Per-fold training and evaluation runtimes are shown in Table 5.6. Total notebook execution time (5 folds, all baselines including LightGCN training) ranges from 5 minutes (Beauty, Fashion) to 30 minutes (Books). The bulk of the runtime is in LightGCN training (PyTorch with BPR loss); the BEST-Rec v4 portion alone takes seconds even on Books.

Dataset	BEST-Rec v4 fit	LightGCN fit	Total all baselines
Beauty	< 1 s	~30 s	~5 min
Fashion	< 1 s	~30 s	~5 min
Instruments	~3 s	~150 s	~12 min
Books	~9 s	~300 s	~30 min

Table 5.6: Per-fold training cost on the warm leave-one-out protocol. BEST-Rec v4’s closed-form solver is two to four orders of magnitude faster than LightGCN’s gradient-based training.

6. Discussion

6.1 Why a Closed-Form Linear Model Beats Deep Recommenders Here

Three structural reasons make EASE+SBERT a particularly strong fit for the small-to-medium Amazon subsets considered. First, the parameter-to-data ratio is favorable: EASE has n^2 parameters in B but they are computed in closed form by inverting a regularized Gram matrix — not learned by SGD — so the “effective” capacity is controlled by λ . Deep models like LightGCN and MultiVAE optimize hundreds of thousands of embedding parameters by gradient descent; on a 2,500-interaction dataset they cannot find a good local optimum without overfitting. Our results in Table 5.2 are consistent with this hypothesis: deep models match or beat us only on the largest dataset (Instruments).

Second, the SBERT content prior provides strong inductive bias exactly where the collaborative signal is weakest: items with few co-occurrences. EASE without this prior (EASE-pure in our tables) underperforms BEST-Rec v4 by 21-47% on the small datasets but only by 2% on Books, where collaborative co-occurrence is already informative. The prior is essentially “free” collaborative information distilled from a model (Sentence-BERT) trained on 1B+ sentence pairs.

Third, EASE’s closed-form solution is rotation-invariant in the latent space, deterministic, and reproducible bit-for-bit across runs. This makes hyperparameter sensitivity easier to analyze (no SGD noise to average across), enables clean ablations (a single matrix change isolates the contribution of every component), and improves engineering practicality (the model can be retrained at every batch insert with negligible cost). Production recommender systems incurring continuous catalog churn benefit substantially from these properties.

6.2 The LC2C Cold-Item Algorithm

LC2C-direct is, to our knowledge, the first cold-item method that learns the per-dataset mapping from textual content to the full collaborative behavior vector of EASE. Two observations from the ablation in Table 5.4 and Figure 5.3 are noteworthy.

First, the SVD-compressed variant (V1) underperforms the direct variant (V2) on every dataset, with the largest gap on Books (+43% V2 over V1). The intuition is that B_{warm} on Books has effective rank well above 64; truncating to $k = 64$ destroys high-frequency components of the collaborative profile. Ridge regression directly to the full n_{warm} -d behavior vector preserves them, while ridge regularization controls overfitting. This finding suggests that future work on content-to-CF mappings should default to the higher-dimensional regression target rather than imposing latent-space compression unnecessarily.

Second, removing the ridge mapping entirely (V3 in Figure 5.3, which uses nearest-warm-in-SBERT as the projector) collapses LC2C’s benefit on Instruments and Books. This confirms that the *learned* mapping is the source of the gain — not the SVD step, not the EASE-warm training, and not the SBERT embeddings on their own. The mapping captures dataset-specific regularities that no fixed similarity (raw cosine, lexical similarity) can express. Concretely, LC2C learns that two items with similar titles in the Books dataset behave differently from two items with similar titles in the Beauty dataset.

6.3 Statistical Significance and Confidence Intervals

Reporting paired-Wilcoxon p-values rather than only point estimates avoids over-claiming tiny gains. For example, on Instruments our model and Higher-Order EASE differ by 0.001 in mean NDCG@10 (Table 5.2), a difference smaller than the per-fold standard deviation. The Wilcoxon test correctly identifies this as not significant ($p = 0.55$), confirming that “adding B^2 ” is statistically equivalent to our content prior on this dataset. Similar caution is required for the cold-user protocol: the per-dataset best method (Table 5.3) varies because of legitimate dataset properties, not because of any one method’s superiority.

6.4 Comparison to the Original BEST-Rec

Compared to the original BEST-Rec architecture (BERT-embedded self-attention transformer with cross-attention fusion), v4 is a substantial simplification: there are no transformer layers, no learned embeddings except the four ridge-regression coefficients, and no gradient-based optimization. The empirical advantage is consistent across all four datasets and across warm, cold-user, and cold-item evaluation protocols. We attribute this gap to the parameter-to-data mismatch noted above; a transformer with hundreds of thousands of parameters cannot be reliably trained on 2,500 to 600,000 interactions, regardless of how well-engineered the cross-attention fusion is. Section 7 acknowledges this design trade-off honestly: BEST-Rec v4 is not designed to scale to billions of interactions where deep recommenders likely re-emerge as the better choice.

7. Limitations and Future Work

BEST-Rec v4 has four notable limitations that suggest avenues for future research.

7.1 Item Catalog Scaling

EASE's matrix inversion is $O(n^3)$ in the item count. Our largest dataset (Books, $n = 13,164$) takes ~ 9 seconds per fold; an industrial catalog with $n \geq 1M$ items would require a different algorithm. Iterative approximations (conjugate gradient, randomized SVD) or block-diagonal partitions of the item space are natural extensions; we have not implemented them in this version.

7.2 Cold-User Beyond Single-Item Context

Our cold-user protocol assumes the user supplies one signup signal — a search query, an interest selection, or an initial review. Pure cold-users (no signal at all) are upper-bounded by popularity, which our results confirm. Production systems with richer user metadata (demographics, device type, geographic context) could integrate that signal as additional input to a hybrid model, which we have not explored here.

7.3 Per-Baseline Hyperparameter Tuning

Each baseline (LightGCN, MultiVAE, iALS, Higher-Order EASE) is run at the hyperparameters reported in its original paper. A more demanding reviewer could insist on a full per-baseline grid search per dataset; our ablation budget did not permit this and we acknowledge it explicitly. We argue that the gap of 2-3x over LightGCN/MultiVAE is large enough that modest per-baseline tuning would not reverse the conclusion. We also note that BEST-Rec v4's own hyperparameters are tuned on 3-fold cross-validation, not on the test fold, eliminating any unfairness in the comparison.

7.4 Sequential Models

Recent recommender benchmarks include sequential models such as SASRec (Kang & McAuley, 2018) and BERT4Rec (Sun et al., 2019). These methods model the temporal sequence of user interactions and predict the next item. Amazon Reviews 2023 timestamps are coarse and incomplete; we therefore did not run a rigorous sequential evaluation. Future work should extend our protocol to a dataset with reliable timestamps (e.g., MovieLens-25M) and compare BEST-Rec v4 against SASRec / BERT4Rec head-to-head.

7.5 Multi-Modal Item Content

Only item titles are used as content. Item descriptions, images (CLIP embeddings), and category structure are also available in the Amazon Reviews 2023 metadata and could be aggregated into a richer S_{content} prior. We have not measured the gain. The design of LC2C is general — it learns a mapping from any d_s -dimensional content vector to the collaborative latent space — so multi-modal extensions require only the construction of an enriched content embedding, not a re-derivation of the algorithm.

8. Conclusion

BEST-Rec v4 is a closed-form linear recommender that augments EASE’s item-item Gram matrix with an SBERT-based content prior, and that introduces LC2C — a learned ridge-regression mapping from textual content to collaborative behavior vectors — for cold items with no training interactions. Across four Amazon Reviews 2023 categories spanning three orders of magnitude in size, BEST-Rec v4 statistically beats six baselines including the most-cited modern deep recommenders (LightGCN, MultiVAE, iALS) on warm leave-one-out NDCG@10, with a 78-113% gain over LightGCN at the extremes. On true cold-item evaluation via GroupKFold-by-item, LC2C-direct achieves 19-141% NDCG@10 improvement over the content-direct content-KNN baseline, with the largest gain on the largest dataset (Books).

The design is deliberately simple: zero learned embeddings, zero gradient steps, four ridge-regression coefficients in the rating head, sub-second training on small datasets, and ~10 seconds on the largest. This simplicity is the source of the model’s robustness: with no overparameterization, BEST-Rec v4 generalizes from extremely sparse training data without the heavy regularization, dropout, or architectural innovations required by deep recommenders. The practical implication is that for small-to-mid Amazon categories, a well-engineered closed-form recommender is the appropriate baseline; the deep alternatives that proliferate in the recommender literature do not yet justify their complexity at this data scale.

All code, preprocessed data caches, hyperparameters, results, ablations, and figures are released as a single Jupyter notebook accompanied by reproducible Python scripts in a uv-managed virtual environment. To reproduce a full evaluation, set DATASET = 'beauty' (or 'fashion', 'instruments', 'books') in the notebook’s config cell and run all cells; expected runtime is 5-30 minutes depending on the dataset.